Davide Sclosa

RESEARCH INTERESTS

DYNAMICAL SYSTEMS, COMBINATORICS, ERGODIC RAMSEY THEORY, GRAPHS AND NETWORKS.

WORK EXPERIENCE

POSTDOC IN MATHEMATICS Nov 2025 - today
King's College London
Strongly coupled maps. With Matteo Tanzi.

POSTDOC IN MATHEMATICS Nov 2024 - Nov 2025
University of Crete
Dynamical Systems with applications in Additive Combinatorics and Analytic Number Theory.
With Nikos Frantzikinakis.

PHD IN MATHEMATICS Nov 2020 - Nov 2024
Vrije Universiteit Amsterdam
Dynamical Systems on Finite Graphs and Graph Limits. With Christian Bick.

EDUCATION

MASTER'S DEGREE IN MATHEMATICS Dec 2018 - Jul 2020
University of Udine.
Full Marks and Honors. Thesis with *Giovanni Panti*.

BACHELOR'S DEGREE IN MATHEMATICS Sep 2015 - Dec 2018
University of Udine.
Full Marks and Honors. Thesis with *Pietro Corvaja*.

PUBLICATIONS

- **MULTIDIMENSIONAL SZEMERÉDI THEOREM FOR PAIRWISE INDEPENDENT JOINTLY INTERSECTIVE POLYNOMIALS**
  with Nikos Frantzikinakis, arXiv preprint arXiv:2610.18657, 2026
- **FROM COMBINATORICS TO GEOMETRY: THE DYNAMICS OF GRAPH GRADIENT DIFFUSION**
  Geometriae Dedicata, vol. 219, n. 1, 2025
- **IF YOU ARE THE SMARTEST PERSON IN THE ROOM, YOU ARE IN THE WRONG ROOM**
  Graphs and Combinatorics, vol 41, n. 5, 2025
- **A RANDOM-WALK CONCENTRATION PRINCIPLE FOR OCCUPANCY PROCESSES ON FINITE GRAPHS**
  with Michel Mandjes and Christian Bick, arXiv preprint arXiv:2410.06807, 2024
- **DIMENSION BOUNDS FOR SYSTEMS OF EQUATIONS WITH GRAPH STRUCTURE**
  with Eddie Nijholt, arXiv preprint arXiv:2410.06840, 2024
- **BOUNDED POWER SERIES ON THE REAL LINE**
  Journal of Mathematical Analysis and Applications, vol. 538, n. 2, 2024
- **DYNAMICAL SYSTEMS ON GRAPH LIMITS AND THEIR SYMMETRIES**
  with Christian Bick, Journal of Dynamics and Differential Equations*, 1-36, 2024*

- **COMPLETELY DEGENERATE EQUILIBRIA OF THE KURAMOTO MODEL ON NETWORKS**
  Nonlinearity, vol. 37, n. 9, 2024
- **KURAMOTO NETWORKS WITH INFINITELY MANY STABLE EQUILIBRIA**
  SIAM Journal on Applied Dynamical SystemsVol. 22, Iss. 4, 2023
- **ALGEBRAIC GROUPS OVER FINITE FIELDS: CONNECTIONS BETWEEN SUBGROUPS AND ISOGENIES**
  Journal of Group Theory, 26 (6), 1143-1155, 2023
- **THE FINITENESS CONJECTURE HOLDS IN** $\mathrm{SL}_2(\mathbb{Z}_{\geq 0})$
  with Giovanni Panti, Nonlinearity, 34 (8), 5234, 2021

CONFERENCES, SEMINARS, SUMMER SCHOOLS

Invited speaker is indicated by ($\star$).

- ***($\star$) 13th Miniworkshop on Operator Theoretic Aspects of Ergodic Theory***,
  CAU Kiel, Germany, 06/10/2026-07/10/2026
- ***Summer School on Additive Combinatorics, Number Theory, and Ergodic Theory***,
  EPFL - Swiss Federal Technology Institute of Lausanne, Switzerland, 14/07/2025-18/07/2025
- ***Perspectives on Ergodic Theory and its Interactions — Celebrating Vitaly Bergelson's Work and Impact***,
  Institute of Mathematics of the Polish Academy of Sciences, Warsaw, Poland, 23/07/2025-27/07/2025
- ***($\star$) Analysis Seminar***,
  University of Crete, Greece, 05/12/2024
- ***($\star$) NetDyn2024 Workshop***,
  Lund University, Sweden, 26/08/2024
- ***Dynamics Days Europe***,
  Bremen, Germany, 29/07/2024-02/08/2024
- ***Hypergraphs: Theory and Applications***,
  The Alan Turing Institute, Isaac Newton Institute, London, UK, 22/07/2024-29/07/2024
- ***($\star$) One World Dynamics***,
  Online, 05/06/2024
- ***($\star$) Networks Seminar***,
  Oxford University, UK, 30/04/2024
- ***($\star$) Dynamical Systems Seminar***,
  Imperial College, London, UK, 26/03/2024
- ***($\star$) Dynamics Seminar***,
  Exeter University, Exeter, UK, 21/03/2024
- ***Floris Takens Seminar***,
  Bernoulli Institute, Groningen, Netherlands, 27/9/23
- ***($\star$) Dynamics Days Europe***,
  Naples, Italy, 03/09/23 - 08/09/23
- ***($\star$) SIAM Dynamical Systems***,
  Portland, US, 15/05/23 - 18/05/23
- ***($\star$) Oberseminar Dynamics***,
  TUM - Technical University of Munich, Munich, Germany, 27/9/21
- ***($\star$) Lowlands Dynamics Seminar***,
  Online, 28/4/2021

TEACHING, MENTORING, AND COORDINATION

- **TEACHING ASSISTANT**
  Tutorials in Dynamical Systems, Discrete Mathematics, and Mathematical Analysis.
- **MENTORSHIP**
  Co-Supervision Bachelor's student.
- **WORKSHOPS ORGANIZER**
  - *Graph Stochastic Dynamics and Dynamic Stochastic Graphs (2024);*
  - *Dynamics of Networks with Switching Topology (2023).*
- **HIRING COMMITTEE**
  Tenure track position (2022), PhD position (2024).